02-Schematic-Review-v1-20032026

Schematic Review Report — Unified_v1 (2026-03-20)

Project: Epicura Unified PCB
Revision: v1
Date Reviewed: 2026-03-20
Source Files: `Unified_v1/` KiCad project (4 sheets: Top, Control, Driver, Connector)
BOM: 87 line items, ~160 components

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1. Architecture Overview

The schematic is organized into 4 hierarchical sheets:

Sheet	File	Contents
Top Level	`Unified_v1.kicad_sch`	Interconnect routing, 3 sub-sheet instances, SWO solder jumper (JP2)
Control	`Control.kicad_sch`	STM32G474RETx MCU, AMS1117-3.3 LDO, crystals, decoupling, boot/reset
Driver	`Driver.kicad_sch`	Buck/boost converters, CAN transceiver, motor drivers, MOSFET array, battery management, I/O expanders, INA219
Connector	`Connector.kicad_sch`	16 external connectors, USB-C + ESD, power input protection, voltage dividers

Inter-sheet wiring: 154 net labels on the top sheet route all signals between sub-sheets. No components on the top sheet except JP2 (SWO jumper).

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2. Power Architecture Summary

Power Rails

Rail	Source	Regulator	Max Capacity	Key Loads
+24V	J_24_VIN1 (XT30, 5A fuse)	Direct	3.2A (PSU limited)	BLDC motor, INA219 sense, buck to 12V
+12V	TPS54531 (U5) from 24V	Buck	3A	Solenoids, fans, linear actuators,

Rail	Source	Regulator	Max Capacity	Key Loads
				pumps
+12V_IN	J_12V_UPS1 (XT30, 5A fuse)	Direct	3A (fuse limited)	UPS backup input to TPS54531
+6.5V	MP1584EN (U8) from 24V	Buck	3A	Currently unloaded (no-connect on top sheet)
+5V_MERGED	Schottky-OR of 3 sources	OR'd	5A (TPS54531 path)	CM5 (up to 3A), LDO, buzzer, LED ring, B0505S
+5V_USB	J_USB1 VBUS (500mA fuse)	Direct	500mA	TP4056 charger, OR'd to 5V_MERGED
+5V_BATT	MT3608 (U14) from Li-Ion	Boost	~500mA	OR'd to 5V_MERGED (micro-UPS)
+5V_ISO	B0505S-1WR3 (U13)	Isolated DC-DC	200mA	ISO1050 VCC2 (CAN bus side)
+3V3	AMS1117-3.3 (U2) from 5V	LDO	800mA	STM32, sensors, I2C pull-ups, ISO1050 VCC1

Three-Source 5V Schottky-OR

```
TPS54531 (5V_MAIN) --[SS14]--> 5V_MERGED
USB VBUS (5V_USB)  --[SS14]--> 5V_MERGED
MT3608 (5V_BATT)   --[SS14]--> 5V_MERGED
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Battery Management

- TP4056 (U16): Li-Ion charger from USB, 1A charge current (R69=1K)
- DW01A (U15) + FS8205A (Q1): Over-charge/discharge/current protection
- MT3608 (U14): Boost 3.7V -> 5V for micro-UPS (RTC/state retention)

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3. MCU Pin Allocation (STM32G474RETx, LQFP-64)

Peripherals

Peripheral	Pins	Function
SPI2	PB12 (NSS), PB13 (SCLK), PB14 (MISO), PB15 (MOSI)	CM5 communication
I2C1	PA15 (SCL), PB7 (SDA)	Sensor bus (5 devices)
FDCAN1	PB8 (RX), PB9 (TX)	Microwave induction CAN
USB FS	PA11 (D-), PA12 (D+)	DFU programming + CDC
SWD	PA13 (SWDIO), PA14 (SWCLK)	Debug
HSE	PF0, PF1	8 MHz crystal
LSE	PC14, PC15	32.768 kHz crystal

Timer / PWM Outputs

Pin	Signal	Timer
PA0	PUMP_PWM	TIM2_CH1
PA6	FAN1_PWM	TIM3_CH1
PA8	BLDC_PWM	TIM1_CH1
PB10	FAN2_PWM	TIM2_CH3
PC3	SLD_OIL_PWM	TIM1_CH4
PC5	SLD_WATER_PWM	TIM1_CH4N (complementary -- see Issue #11)
PC7	BUZZER_PWM	TIM3_CH2

GPIO

Pin	Signal	Direction
PA1	24V_Fail_Detection (COMP1)	Input (analog)
PA3	ADC_BUS_Detection	Input (analog)
PA4	BLDC_EN	Output
PA5	BLDC_DIR	Output
PA10	CID1_Actuator_EN	Output
PB0	SAFETY_RELAY	Output
PB1	REED_SWITCH	Input
PB2	E-STOP	Input (EXTI)
PB3	IRQ_CM5 (shared SWO via JP2)	Output

Pin	Signal	Direction
PB4	CID1_Actuator_DIR	Output
PB5	CID2_Actuator_EN	Output
PB6	CID1_HOME	Input
PC2	CID2_Actuator_DIR	Output
PC4	SLD_OIL_DIR	Output
PC6	SLD_WATER_DIR	Output
PC9	HX711_SCK_OIL	Output
PC10	HX711_DOUT_OIL	Input
PC11	HX711_SCK_Water	Output
PC12	HX711_DOUT_Water	Input
PD2	CID2_HOME	Input

Unused Pins (8 pins, all No-Connect)

PA2, PA7, PA9, PB11, PC0, PC1, PC8, PC13

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4. Connector Summary (16 connectors)

Ref	Type	Pins	Purpose	Protection
J_24_VIN1	XT30PW-M	2	24V PSU input	5A PTC + SS54 reverse + SMBJ24A TVS
J_12V_UPS1	XT30PW-M	2	12V UPS input	5A PTC + SS54 reverse + SMBJ24A TVS
J_USB1	USB-C 16P	16	DFU + CDC + charging	500mA PTC + SS54 + USBLC6-2SC6 ESD
J_CM5_1	2x20 2.54mm	40	CM5IO stacking (SPI2, IRQ, 5V, GND)	None
J_BLDC1	DF11-6DP	6	BLDC motor (24V, PWM, EN, DIR)	None
J_CAN1	B4B-XH	4	CAN bus (CAN_H, CAN_L, GND_ISO)	Isolated ground only
J_PASD1	DF11-16DP	16	7x solenoid drives + 12V + GND	None
J_SLD1	DF11-16DP	16	2x pump, 2x solenoid, 2x HX711	None

Ref	Type	Pins	Purpose	Protection
J_CID1	DF11-14DP	14	Stepper diff + actuator + limits + 24V	100R series on stepper
J_FAN1	B4B-XH	4	2x fan (12V + drain)	None
J_IR_LED1	DF11-8DP	8	I2C (MLX90614) + LED ring (WS2812B)	None
J_PRES1	DF11-4DP	4	I2C (ADS1015 pressure)	None
J_SAFE1	B4B-XH	4	Safety relay, reed switch, E-stop	None
J_SWD1	TC2030	6	SWD debug (TagConnect)	None
J_BAT1	B2B-PH	2	Li-Ion battery	None
J_BUZZ1	B2B-PH	2	5V buzzer	None

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5. Issues Found

5.1 Critical Issues (must fix before fabrication)

CRIT-1: TPS54531 feedback divider gives ~5V instead of 12V

Sheet: Driver | **Components:** U5, R20 (52.3K), R67 (10K)

The feedback divider calculates: Vout = 0.8V x (1 + 52.3K/10K) = **4.98V**, not 12V. The hierarchical label confirms 12V is intended (+12V_Rail). For 12V output, the upper resistor should be ~140K with 10K lower: Vout = 0.8 x (1 + 140/10) = 12V.

Impact: 12V rail will regulate to ~5V. All solenoids, fans, linear actuators, and pumps will be underpowered or non-functional. DRV8876 and TB6612FNG motor drivers will not operate correctly.

Fix: Change R20 from 52.3K to 140K.

CRIT-2: MT3608 feedback divider gives ~6.6V instead of 5V

Sheet: Driver | **Components:** U14, R23 (100K), R32 (10K)

The feedback divider calculates: Vout = 0.6V x (1 + 100K/10K) = **6.6V**, not 5V. For 5V output with 10K lower, the upper resistor should be ~73.3K.

Impact: Battery boost output will be 6.6V, exceeding the 5.25V absolute max of many 5V-rated components on the 5V_MERGED bus. Could damage the CM5 or STM32 LDO.

Fix: Change R23 from 100K to 73.3K (nearest standard: 73.2K or 75K).

CRIT-3: SI2301-TP (P-channel) uses 2N7002 (N-channel) symbol

Sheet: Driver | **Component:** Q2

The symbol is `Transistor_FET:2N7002` (N-channel MOSFET) but the value is `SI2301-TP` (P-channel MOSFET). Gate, source, and drain pins are mapped incorrectly in the netlist. The circuit will not function as designed.

Impact: The P-FET switching circuit will have reversed source/drain and wrong gate polarity. The generated netlist and PCB layout will have incorrect connections.

Fix: Replace the 2N7002 symbol with a proper P-MOSFET symbol (e.g., `Transistor_FET:SI2301CDS`) and verify pin mapping.

CRIT-4: 12V input TVS clamp voltage too high (39.2V)

Sheet: Connector | **Component:** D_IN3 (SMBJ24A)

The 12V UPS input uses an SMBJ24A TVS with 39.2V clamping voltage. Downstream 12V components will see up to 39V before the TVS conducts -- this exceeds the ratings of 12V-rated components.

Impact: TVS provides no effective protection for the 12V rail. A transient could destroy the TPS54531 (max VIN = 40V, marginal) and downstream components.

Fix: Change D_IN3 from SMBJ24A to SMBJ12A (13.3V standoff, 19.9V clamping).

CRIT-5: J_SAFE1 symbol-footprint mismatch

Sheet: Connector | **Component:** J_SAFE1

The schematic symbol is B4B-XH-A (JST XH, single-row 4-pin, 2.5mm pitch) but the footprint is CON4_2X2_RF_DF11_HIR (Hirose DF11, dual-row 2x2, 2.0mm pitch). These are physically incompatible connectors with different pin counts per row.

Impact: BOM will list wrong part. Pin mapping between single-row (1-2-3-4) and dual-row (1-2 / 3-4) may result in swapped safety signals. Safety relay, E-stop, and pot detection could be wired incorrectly.

Fix: Change the symbol to match the intended connector (likely DF11-4DP-2DS to match the footprint), and verify pin assignments.

5.2 Major Issues (strongly recommended to fix)

MAJ-1: No TVS/ESD protection on CAN bus at connector

Sheet: Connector | **Component:** J_CAN1

CAN_H and CAN_L have no TVS diodes at the connector entry point. The CAN bus cable runs to the microwave induction module -- a high-EMI environment with switching transients from the induction coil.

Fix: Add PESD2CAN (or CDSOT23-SM712) TVS diode across CAN_H/CAN_L at J_CAN1.

MAJ-2: VREF+ has no dedicated filtering

Sheet: Control | **Component:** Pin 28 (VREF+)

VREF+ is connected directly to +3V3 without a dedicated ferrite bead or decoupling capacitor pair. ST application notes recommend 1uF + 100nF on VREF+.

Impact: ADC readings on PA1 (24V detection) and PA3 (bus detection) will be noisy, potentially causing false power-failure triggers.

Fix: Add a ferrite bead and 1uF + 100nF decoupling on VREF+, separate from the VDDA filtering.

MAJ-3: VDDA missing bulk capacitor

Sheet: Control | **Component:** FB1, C9

VDDA has ferrite bead FB1 (600R@100MHz) and a single 100nF cap (C9). ST AN4206 recommends 1uF + 100nF on VDDA.

Fix: Add a 1uF ceramic capacitor in parallel with C9 on VDDA.

MAJ-4: No UART fallback for CM5-STM32 communication

Sheet: Control

The design doc specifies "SPI (primary) or UART (fallback)" but no UART pins are assigned. PA2 and PA9 (capable of USART2_TX and USART1_TX) are both No-Connect.

Fix: Route at least one UART (TX+RX) to J_CM5 or to test points. PA2 (USART2_TX) and PA3 (USART2_RX) are available, though PA3 is currently used for ADC.

MAJ-5: TB6612FNG VM bypass capacitor undersized

Sheet: Driver | **Component:** C_VM2 (10uF)

The TB6612FNG motor power bypass is only 10uF. The datasheet recommends 100uF+ for motor applications with inductive loads (peristaltic pumps).

Impact: Voltage sag during pump startup could cause erratic motor behavior or brownout the 12V rail.

Fix: Increase C_VM2 to 100uF (electrolytic) or add a 100uF capacitor in parallel.

MAJ-6: Missing input bulk capacitor on TPS54531

Sheet: Driver | **Component:** U5

The TPS54531 datasheet requires a minimum 20uF ceramic input capacitor close to VIN. Only C35 (100nF) is visible as VIN bypass. The 100uF (C21) is on the output side.

Impact: Input voltage ringing, potential instability, and EMI issues on the 24V-to-12V converter.

Fix: Add a 22uF ceramic capacitor (25V rated, 0805 or 1206) on U5 VIN, placed as close as possible to the IC.

MAJ-7: No 12V rail voltage monitoring

Sheet: Connector

The 24V rail has a voltage divider (R1/R9) for ADC monitoring, but the 12V rail has no equivalent. The 12V rail operates at 0% headroom under peak load and powers critical actuators.

Fix: Add a voltage divider on the 12V rail to an available ADC pin (e.g., PA7 or PC0).

MAJ-8: No protection on BLDC control lines

Sheet: Connector | **Component:** J_BLDC1

BLDC_PWM, BLDC_DIR, BLDC_EN are routed directly to the connector with no series resistors or ESD protection. The BLDC motor/ESC environment generates significant EMI.

Fix: Add 100R series resistors on all three signal lines, and consider a TVS array.

MAJ-9: No ESD protection on external I2C lines

Sheet: Connector | **Components:** J_IR_LED1, J_PRES1

I2C_SCL and I2C_SDA run over cables to external boards (MLX90614, ADS1015) with no ESD protection at the connector.

Fix: Add an I2C-rated ESD protection device (e.g., USBLC6-4SC6 or IP4220CZ6) near J_IR_LED1.

MAJ-10: No filtering on HX711 load cell signals

Sheet: Connector | **Component:** J_SLD1

HX711_OIL and HX711_WATER clock/data lines run over cables with no series resistors or decoupling. The HX711 is a 24-bit ADC extremely sensitive to noise.

Fix: Add 100R series resistors and 100pF caps to GND on each HX711 signal line at the connector.

5.3 Minor Issues (recommended)

MIN-1: INA219 label naming inconsistency

Sheet: Driver | **Component:** U7

The hierarchical label reads `+24V_INA226_IN` but the component is an INA219 (U7), not INA226. Creates confusion during firmware development.

Fix: Rename label to `+24V_INA219_IN`.

MIN-2: LED current limiting resistors too high

Sheet: Driver | **Components:** R75, R67 (10K each)

At 3.3V with ~2V LED Vf: I = (3.3 - 2) / 10K = 0.13mA. The charging (D23) and standby (D22) LEDs will be invisible.

Fix: Change R75 and R67 to 1K (1.3mA) or 2.2K (0.6mA).

MIN-3: MP1584EN FREQ resistor out of range

Sheet: Driver | **Component:** U4, R64 (732K)

The MP1584EN datasheet specifies FREQ resistor range of 100K-360K (200kHz-1.5MHz). R64 = 732K is well outside this range.

Fix: Change R64 to a value within 100K-360K per the datasheet frequency table.

MIN-4: No debounce on switch inputs

Sheet: Control | **Pins:** PB1 (reed switch), PB2 (E-stop), PB6 (CID1 home), PD2 (CID2 home)

No hardware RC debounce networks on any mechanical switch inputs. Software debouncing is possible but hardware debounce is more reliable for safety-critical inputs like E-stop.

Fix: Add RC debounce (10K + 100nF = 1ms time constant) at minimum on PB2 (E-stop). Consider for all switch inputs.

MIN-5: Boot0 pin connection unclear

Sheet: Control

A text note reads "Where to connect Boot) PB* is already used for something else." The Boot0 circuit (SW1 + R4 pull-down) exists but its connection to the MCU's BOOT0 pin is not clearly traceable.

Fix: Verify BOOT0 pin is properly connected and remove the ambiguous text note.

MIN-6: PC5 timer conflict for SLD_WATER_PWM

Sheet: Control | **Pin:** PC5

PC5 can use TIM1_CH4N (complementary output). Since TIM1_CH4 is already used on PC3 for SLD_OIL_PWM, CH4 and CH4N are complementary pairs of the same channel --

they cannot be independently controlled.

Fix: Move SLD_WATER_PWM to a pin with an independent timer channel (e.g., PC8/TIM3_CH3 or PB11/TIM2_CH4, both currently NC).

MIN-7: PCF8574AT address conflict risk

Sheet: Driver | **Components:** U9, U10

Both I/O expanders are PCF8574AT (base address 0x38). If A0/A1/A2 pins are identically configured, they will collide on the I2C bus.

Fix: Verify A0/A1/A2 pin states differ between U9 and U10.

MIN-8: No bulk VDD capacitor near MCU

Sheet: Control

Four VDD pins have 100nF decoupling each, but no single bulk capacitor (4.7-10uF) is placed near the MCU. The 10uF caps at the LDO output may be physically distant on the PCB.

Fix: Add a 4.7uF ceramic capacitor near the MCU VDD cluster.

MIN-9: Unused GPIO pins not routed to test points

Sheet: Control | **Pins:** PA2, PA7, PA9, PB11, PC0, PC1, PC8, PC13

Eight GPIO pins are No-Connect. For a prototype, routing these to test points or a header would provide debug flexibility and future expansion.

Fix: Route unused pins to test pads or a 0.1" header.

MIN-10: J_PASD1 pin 15 is No-Connect instead of GND

Sheet: Connector | **Component:** J_PASD1

Pins 11-14 and 16 are all GND, but pin 15 has a no-connect flag. This wastes a ground return pin on a 16-pin connector carrying 7 solenoid drives.

Fix: Connect pin 15 to GND.

MIN-11: GPIO18 uses local label instead of hierarchical

Sheet: Connector

GPIO18 (LED ring data from CM5) is connected between J_IR_LED1 and J_CM5 via a local label instead of a hierarchical label. Inconsistent with all other inter-sheet signals.

Fix: Change to hierarchical label for consistency.

MIN-12: USB CC configuration is device-only

Sheet: Connector | **Components:** R13, R14 (5.1K to GND)

The 5.1K pull-downs on CC1/CC2 configure USB-C as UFP (device/sink only). This is correct for DFU programming and charging, but prevents USB host mode if ever needed.

Action: Confirm this matches design intent (likely correct for this application).

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6. Issue Summary

Severity	Count	IDs
Critical	5	CRIT-1 through CRIT-5
Major	10	MAJ-1 through MAJ-10
Minor	12	MIN-1 through MIN-12
Total	**27**	

Priority Fix Order

1. **CRIT-1** (TPS54531 feedback) and **CRIT-2** (MT3608 feedback) -- wrong output voltages, simple resistor value changes
2. **CRIT-3** (SI2301 wrong symbol) -- incorrect netlist generation
3. **CRIT-4** (12V TVS value) -- no effective protection on 12V rail
4. **CRIT-5** (J_SAFE1 mismatch) -- safety connector pin mapping error
5. **MAJ-1** (CAN TVS) -- EMI protection for safety-critical interface
6. **MAJ-2/3** (VREF+/VDDA filtering) -- ADC accuracy for safety monitoring
7. **MAJ-6** (TPS54531 input cap) -- converter stability
8. Remaining major and minor issues

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7. BOM Observations

- **Total unique parts:** 87 line items
- **Total components:** ~160 placed
- **Key ICs:** STM32G474RETx, 2x MP1584EN, TPS54531, MT3608, AMS1117-3.3, ISO1050DUB, B0505S-1WR3, INA219, DRV8876, TB6612FNG, 2x PCF8574AT, TP4056, DW01A, FS8205A, USBLC6-2SC6
- **MOSFET switches:** 12x IRLML6344 + 1x 2N7002 (+ Q2 SI2301 P-FET)
- **Protection:** 3x SS54, 3x SS34, 15x SS14, 2x SMBJ12A, 2x SMBJ24A
- **Fuses:** 2x 5A PTC (power inputs), 2x 1.5A (battery path), 1x 500mA (USB)
- **Connectors:** 16 total (2x XT30, 1x USB-C, 1x 2x20 header, 1x TagConnect, 6x Hirose DF11, 3x JST-XH, 2x JST-PH)

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8. Cross-Reference: Schematic vs Design Document

Specification (from 01-Unified-PCB-Design.md)	Schematic Status
STM32G474RETx LQFP-64, all 48 GPIO allocated	40 used, 8 NC -- matches (8 freed to PCF8574)
SPI2 for CM5 communication	PB12-PB15 correctly assigned
FDCAN1 for microwave induction	PB8-PB9 correctly assigned
I2C1 with 4.7K pull-ups	PA15, PB7 with R2/R3 = 4.7K -- correct
24V -> 12V via TPS54531	Present but **feedback resistor wrong (CRIT-1)**
24V -> 6.5V via MP1584EN	Present, output is no-connect (unused)
Battery -> 5V via MT3608	Present but **feedback resistor wrong (CRIT-2)**
B0505S-1WR3 for CAN isolation	Present with correct isolation domains
DRV8876 for CID linear actuators	Present with proper H-bridge connections
TB6612FNG for peristaltic pumps	Present but **VM cap undersized (MAJ-5)**
2x PCF8574AT I/O expanders	Present, verify address differentiation
INA219 current monitor on 24V	Present, 10m shunt, naming inconsistency
TP4056 + DW01A + FS8205A battery management	Present and correctly wired
USBLC6-2SC6 USB ESD	Present on USB D+/D-
CAN isolation slot 6mm	Layout concern -- not verifiable from schematic

Specification (from 01-Unified-PCB-Design.md)	Schematic Status
UART fallback for CM5	**Not implemented (MAJ-4)**